

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Image sensor with industry's smallest 0.5µm pixel

The ISOCELL HP3 from Samsung is a 200MP camera sensor with the world's smallest sensor measuring only 0.5µm. The sensor packs 200 million pixels in a 1/3.6cm optical format. ISOCELL HP3 is capable of capturing HDR images with 14-bit RAW output and employs a pixel



binning technique called Tetra2pixel technology, which joins up to 16 pixels to capture a 12MP image with much bigger pixels in an ultra-low light condition. All the pixels are equipped with AF capabilities, which allow for sharper and more vibrant images. Furthermore, it allows more accurate and quicker auto-focus. The sensor is capable of shooting 8K videos @ 30fps and 4K videos @ 120fps. The smaller pixels have resulted in a 20% reduction in sensor size compared to its predecessor, thus making it desirable for high-end smartphones.

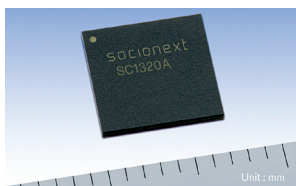
Samsung

<https://semiconductor.samsung.com>

World's first HD-PLC LSI compliant with IEEE 1901-2020

The SC1320A HD-PLC communication LSI from Socionext is the world's first semiconductor design using an IEEE 1901-2020 compliant IP core. The High-Definition Power Line Communication (HD-PLC) superimposes high frequency signals over existing power lines,

telephone lines, etc, to enable communication. The SC1320A uses only a single 3.3V power supply and comes in a compact 7 × 7mm package. This solution has multiple I/O interfaces,



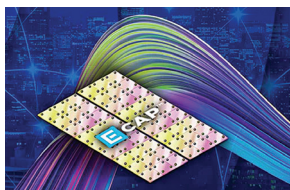
such as SPI/UART and Ethernet MAC (RMII), allowing SC1320A to be easily embedded into designs. The compact and low-power solution is suitable for a wide variety of applications, which include home automation and smart metering.

Socionext

<https://www.socionext.com>

World's smallest and most configurable capacitor

The latest addition to E-Cap series of capacitor solution from Empower Semiconductor is the industry's high-performance, smallest size, and



most configurable capacitor. E-CAP solutions offer densities of 1.1µF/mm², which is over twice the density of alternative silicon capacitor technologies. E-Cap provides a superior high-frequency de-coupling solution with a much smaller footprint and component count than traditional MLCC based solutions. It is suitable for use in demanding applications, such as IoT, wearables, mobile, and processors

where size, performance, and flexibility are essential.

Empower

<https://www.empowersemi.com>

Energy-efficient Bluetooth location service device

BG22 from Silicon Labs is an energy-efficient Bluetooth location service solution. It incorporates advanced software, which is designed to maximise the location-finding capabilities of the BG22 series of SoCs and



SiP modules. The device features an accurate, low-power Bluetooth device to simplify angle of arrival (AoA) and angle of departure (AoD) location services. This product family combines ultra-low transmit and receive power (4.1mA TX at 0dBm, 3.6mA RX) with a high-performance Arm Cortex-M33 core (27µA/MHz active, 1.2µA sleep). It can operate for up to ten years on a single coin cell battery and can be used in applications such as asset tracking, warehouse management, IoT for healthcare products, and more through real-time location services (RTLS).

Silicon Labs

<https://www.silabs.com>

Ultra-accurate GNSS module to cover all major bands

The ORG4600 from Origin GPS is a dual-frequency L1 + L5 GNSS for ultra-accurate positioning. The highly